

"Falcon 2.0T: Timing chain stretch sets crankshaft/camshaft correlation fault"

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Area: bulletins

1 Condition

Falcon 2.0T vehicles may store diagnostic trouble code P0016 (crankshaft position/camshaft position correlation, bank 1 sensor A) and may also set P0011 (camshaft position timing over-advanced or system performance, bank 1). Symptoms include a hard or extended crank, rough cold start, reduced power, and an illuminated malfunction indicator lamp. P0016 indicates the measured crank-to-cam phase has drifted outside the correlation window. Begin diagnosis using the Falcon Engine Service Manual :: Timing System .

2 Cause

The primary cause is mechanical stretch of the timing chain TC-3340, which retards camshaft timing relative to the crankshaft and sets P0016. As the chain elongates beyond its wear limit, the variable valve timing actuator VVT-3360 can no longer achieve commanded phase, which may additionally set P0011. Inspect the chain guides and tensioner condition before condemning related actuators. Phasing operation is described in the Falcon Engine Service Manual :: Variable Valve Timing System .

3 Repair Procedure

Perform PROC-CPS-CMP to verify crankshaft and camshaft position alignment, then replace the stretched timing chain TC-3340 along with its guides and tensioner per the published service procedure. If the VVT-3360 solenoid fails to hold commanded position after the chain is renewed, replace it as well. Chain renewal steps are in the Falcon Engine Service Manual :: Timing Chain Service , and solenoid replacement is in the Falcon Engine Service Manual :: VVT Solenoid Replacement . The closely related Harrier platform shares this architecture; equivalent steps are in the Harrier Engine Service Manual :: Timing Chain Service and the Harrier Engine Service Manual :: Variable Valve Timing Solenoid Replacement . After repair, clear P0016 and P0011 and confirm the codes do not return after several drive cycles.